

Homework 13: General Knowledge Sweep I

Paper track. This one is not tied to a single session — it covers the loose short-answer items that show up in the "general knowledge / junior detective" section of real invitational tests. Use the Knowledge Document to check yourself. No home lab.

Part A: Powders — uses as identification clues (1 pt each)

Tests often give the **common use** of a powder as a clue. Name the powder (and formula) from its use:

1. Making plaster / casts
2. Antacid / "stomach aches" (fizzes in water)
3. Chalk; also an antacid and a building material
4. Thickening agent (turns iodine black)
5. "Helps with the common cold" (decolorizes iodine)
6. Flavor enhancer / umami
7. Making concrete
8. Making glass
9. Jello / gelatin desserts
10. Bread (tan pellets, biological)

Part B: Flame-test colors (1 pt each)

A flame test can help identify a metal or metal salt. Give the flame color:

1. Copper
2. Sodium (or table salt contamination)
3. Potassium
4. Calcium
5. Lithium / strontium
6. Which two of the six Crime Busters metals are hardest to tell apart by flame color alone, so you'd use density or acid reaction instead?

Part C: Natural vs. synthetic polymers (1 pt each)

1. Name three **naturally occurring** polymers.
2. Name three **synthetic** polymers.
3. Is rayon natural, synthetic, or "regenerated"? Explain.
4. Under a microscope, how do natural fibers usually look different from synthetic ones?

Part D: DNA & PCR (2 pts each)

1. How many chromosomes does a human have? How many pairs?
2. What does RNA stand for, and name one way it differs from DNA.
3. What are the three stages of PCR, in order?
4. What enzyme adds nucleotides to the growing DNA strand during PCR?
5. Why is PCR done *before* gel electrophoresis?
6. On a gel, do longer or shorter fragments travel farther? Why?

Part E: Blood spatter — the information list (5 pts)

"What can you learn from a blood spatter pattern?" List **six** things.

Part F: Blood spatter — drops & velocity (1 pt each)

1. As the height a blood drop falls from **increases**, what happens to the diameter of the stain (up to a point)?
2. Give the rough drop-size and cause for **low**, **medium**, and **high** velocity spatter.
3. A stain 5 mm wide and 10 mm long — angle of impact?

Part G: Minutiae (1 pt each)

Name the minutia from the description:

1. A single ridge splits into two.
2. A ridge starts, then stops, with no connection.
3. A very short ridge, longer than a dot, between two others.
4. A ridge that connects two parallel ridges.
5. A single ridge divides and comes back together, leaving a gap.
6. The triangular formation where ridge flows meet.

Part H: Odds and ends (1 pt each)

1. State Locard's Exchange Principle.
2. What two body structures (besides hair) are also made of keratin?
3. What hormone group regulates human hair growth?
4. What is the study of fingerprints called? Of handwriting?
5. Define "agglutination" as used in blood typing.
6. Name three types of chromatography.
7. In paper chromatography, what two forces make the solvent climb the paper?
8. What is the empirical formula of acetic acid ($C_2H_4O_2$)?
9. How many note sheets may each participant bring, and what size?
10. List five things that can be determined from a shoeprint.

Answer Key

Part A: 1. Gypsum / plaster of Paris — $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ ($\frac{1}{2}\text{H}_2\text{O}$ when calcined). 2. Alka-Seltzer. 3. Calcium carbonate — CaCO_3 . 4. Cornstarch — $(\text{C}_6\text{H}_{10}\text{O}_5)_n$ (or flour). 5. Vitamin C (ascorbic acid) — $\text{C}_6\text{H}_8\text{O}_6$. 6. MSG — $\text{C}_5\text{H}_8\text{NO}_4\text{Na}$. 7. Sand — SiO_2 (or gypsum/Portland components). 8. Sand (silica) — SiO_2 . 9. Gelatin. 10. Yeast.

Part B: 1. Green / blue-green. 2. Bright yellow-orange. 3. Lilac / pale violet. 4. Orange-red (brick red). 5. Red / crimson. 6. Any of magnesium/aluminum/zinc/tin/iron give a weak or white-ish flame with no distinctive color — use density, magnetism (iron), color (copper), and acid vigor instead. (Copper's green is the one clearly diagnostic flame in the set.)

Part C:

1. Any three: cellulose (cotton, wood), starch, cellulose/lignin, natural rubber (latex), silk (fibroin), wool/hair (keratin), DNA, proteins.
2. Any three: polyethylene, polypropylene, polystyrene, PVC, PET, nylon, polyester, PMMA, polycarbonate, Bakelite.
3. Regenerated / semi-synthetic — it's made by chemically processing natural cellulose from wood pulp, so it starts natural but is manufactured. (Some tests just call it "synthetic.")
4. Natural fibers are irregular — uneven thickness, twists, imperfections, scales (wool) or ribbon shape (cotton); synthetic fibers are smooth and uniform in diameter, often with extrusion lines and delustering pits.

Part D:

1. 46 chromosomes; 23 pairs.
2. Ribonucleic acid; single-stranded (vs. double), ribose sugar (vs. deoxyribose), uracil replaces thymine.
3. Denaturation (strands separate), annealing (primers bind), extension / synthesis (new strand built).
4. DNA polymerase (Taq polymerase).
5. PCR amplifies the tiny DNA sample into millions of copies so there's enough to see as visible bands on the gel.
6. Shorter fragments travel farther — they move through the gel's pores more easily; DNA's negative charge pulls all fragments toward the positive electrode.

Part E: Any six of: direction of travel; angle of impact; the velocity / force involved (low/medium/high); the likely cause (drip, blunt force, gunshot, cast-off, arterial); the area of convergence and area of origin (where the blood came from); the position and movement of the victim/attacker; the number of blows (from cast-off count); whether someone/something was present (a void); blood type; DNA; sequence/timeline of events; whether the scene was altered.

Part F:

1. The stain diameter **increases** as drop height increases, until it reaches a terminal size (around 7 feet) and stops growing.

2. Low: large drops > 4 mm, gravity only (dripping wound, pooling); medium: 1–4 mm, blunt-force/stabbing; high: fine mist < 1 mm, gunshot/explosion.
3. $\arcsin(5 \div 10) = \arcsin(0.5) = 30^\circ$.

Part G: 1. Bifurcation. 2. Ridge ending. 3. Island / short ridge. 4. Bridge. 5. Lake / enclosure. 6. Delta.

Part H:

1. Every contact leaves a trace — material transfers both ways when two objects touch.
2. Fingernails and the outer layer of skin (epidermis). (Also accept: hooves, horns, feathers, claws.)
3. Androgens.
4. Fingerprints: dactyloscopy. Handwriting: graphology (or questioned-document examination).
5. The clumping of red blood cells that occurs when an antibody meets its matching antigen — how blood type is read from a test.
6. Any three: paper, thin-layer (TLC), column, gas (GC), high-performance liquid (HPLC).
7. Adhesion (water to paper) and cohesion (water to water) — together, capillary action.
8. CH_2O .
9. One 8.5" × 11" sheet per participant, both sides, in a sheet protector.
10. Any five: direction of travel; walking vs. running (from stride length and toe depth); approximate shoe size; the shoe's tread pattern / type / brand; number of people; an injury or unusual gait; approximate weight (from depth in a 3-D print); individual defects that identify one specific shoe.